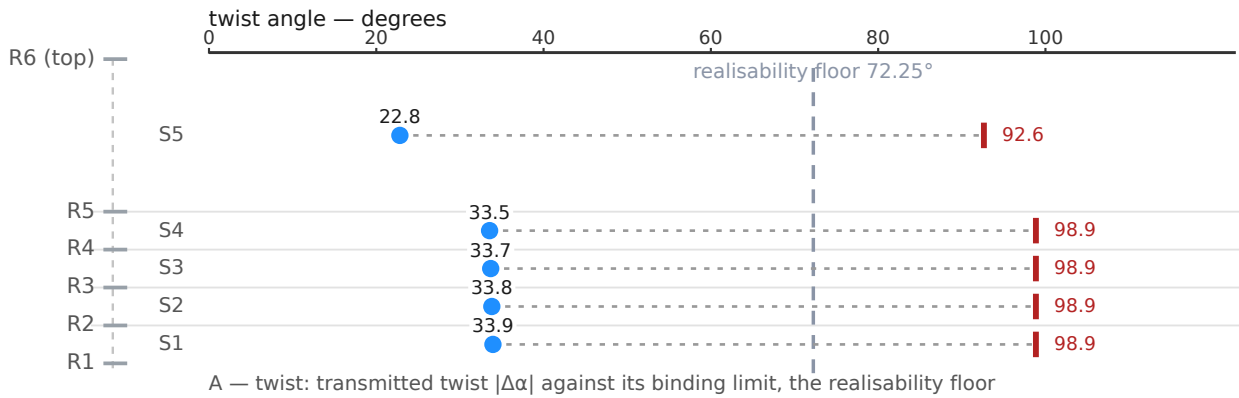
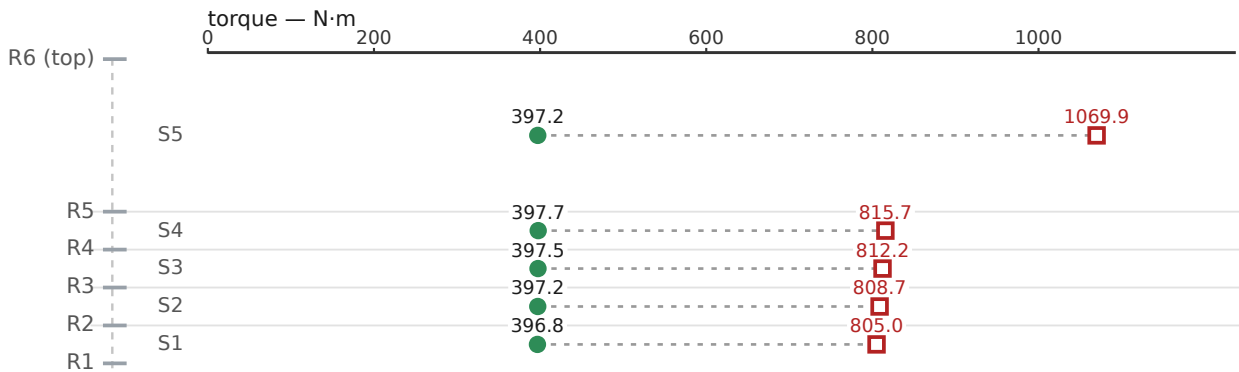
Every segment of the winner runs well inside its binding limit

Transmitted twist against its binding limit, the realisability floor · shaft profile S1–S5, ground to top · island-3 winner operating point



A — twist: transmitted twist |Δα| against its binding limit, the realisability floor



B — torque: segment torque against torsional capacity, paired at the same state

transmitted twist (panel A) · segment torque (panel B) · δcrit over-twist limit · torsional capacity

realisability floor — sin Δα ≤ 1/1.05 holds the geometry only to 72.25°.

crossing limit δα* — every segment gates on its own collapse-detector limit. That limit is not the over-twist limit.

no limit — the segment's tether is shorter than the sum of its end radii. Its lines cannot reach the axis,

so no over-twist limit exists. Tether break strain and ring compression carry the limit instead.

source: data commit dd3cc6a · extract extract-winner-operating-point.csv · capture pair-extract-2026-10-08-winner-ops

s = 1724859 (t = 49.99999483862295 s) · rows NR-022 · NR-023 · script commit 1af63fc · authority commit f8114be